Chain-of-Cooking: Cooking Process Visualization via Bidirectional Chain-of-Thought Guidance

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Abstract

Cooking process visualization aims to generate a coherent sequence of images depicting each procedural step of a recipe, while faithfully tracking evolving ingredient appearances and maintaining stepwise ordering. Existing approaches focus primarily on final-dish generation or single-step image synthesis and suffer from semantic drift and incoherent sequences when applied to multi-step procedural generation. To address these challenges, we propose **Chain-of-Cooking (CoCook)**, a diffusion-based framework that unifies dynamic visual evidence retrieval with bidirectional chain-of-thought (CoT) prompt evolution during the denoising process. CoCook comprises three specially designed modules: (1) a *Dynamic Patch Selection* (DPS) module that retrieves text-matched patches from historical frames and injects them into the diffusion process via a learnable visual attention path; (2) a *Semantic Evolution* (SE) module that initializes latent prompts from the previous step's text embedding and cross-attends with the current step text to capture step-aware contextual cues; and (3) a *Bidirectional CoT Guidance* mechanism that updates latent prompts in both forward and backward denoising passes with a semantic-consistency loss to preserve procedural ordering. Furthermore, we introduce **CookViz**, a new public benchmark of 40,362 high-resolution, step-aligned image–text pairs in English for evaluating cooking process visualization. Comprehensive experiments on both RecipeQA and CookViz demonstrate that CoCook outperforms all baselines, achieving FID scores of 12.43 and 12.28 respectively, and yielding the highest scores in human evaluations for semantic consistency and frame coherence.

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1 Introduction

Food image generation has attracted substantial research interest due to its broad applications in dietary analysis, culinary recommendation, and nutritional monitoring [5, 12, 14, 17, 26, 31, 32]. Early methods focus on generating a single, plausible food image conditioned on recipe text or ingredient lists [31, 32]. However, real-world cooking is inherently procedural: a recipe unfolds across multiple ordered steps, each transforming raw or partially prepared ingredients into a new visual state. *Cooking process visualization*—the task of generating one faithful image per recipe step while maintaining appearance fidelity and procedural coherence—remains largely unexplored.

As illustrated in Fig. 1, traditional food image generation (left) produces a single output depicting the final dish, whereas cooking process visualization (right) requires synthesizing a coherent sequence of intermediate states—from raw ingredients through successive transformations to the finished product. This sequential nature introduces two key challenges: (i) **appearance fidelity**, i.e., ingredients and cookware must evolve consistently across frames; and (ii) **stepwise coherence**, i.e., the ordering of visual events must align with the procedural logic of the recipe.

Existing generative models struggle with both challenges. Standard diffusion models such as Stable Diffusion [22] and SDXL [18] independently condition each image on its corresponding step text, ignoring historical visual context and thus producing semantically inconsistent sequences. Story generation methods like StoryGen [11] and StoryLDM [20] model narrative coherence for general scenes but are not designed for the fine-grained ingredient-level transformations of procedural cooking. Retrieval-augmented methods such as ImageRAG [23] and TIP [13] incorporate reference images but do not exploit temporal ordering or step-to-step semantic evolution.

A further bottleneck is the lack of suitable benchmarks. While datasets such as Cookpad [6] and MIAIS [29] contain cooking images, they are private or in Japanese; RecipeQA [27] is public and English but contains only approximately 20,000 frames and lacks explicit step-level image alignment for generation evaluation.

To overcome these limitations, we make the following contributions:

- We propose **CoCook**, a diffusion-based cooking process visualization framework with three novel modules: Dynamic Patch Selection (DPS), Semantic Evolution (SE), and Bidirectional Chain-of-Thought (CoT) Guidance, built on a frozen Stable Diffusion v1.5 backbone.
- We introduce **CookViz**, a public benchmark of 40,362 high-resolution, step-aligned English image–text pairs with at

least three steps per recipe, designed specifically for evaluating multi-step cooking process visualization.
- We propose a training objective combining denoising loss and semantic-consistency loss to align evolving latent prompts with current step semantics across both forward and backward denoising directions.
- We conduct comprehensive experiments on RecipeQA and CookViz with automatic metrics (FID, CLIP-T, CLIP-I, DreamSim) and a 50-participant human study, demonstrating state-of-the-art performance across all metrics.

2 Related Work

2.1 Food Image Generation

Food image generation aims to produce realistic food images conditioned on recipe descriptions or ingredient lists. Early GAN-based methods [12, 17, 31, 32] encode recipe text and ingredients jointly to condition an adversarial generator. CookGALIP [26] leverages CLIP-based alignment and sequential ingredient prompts to improve semantic consistency. More recently, diffusion-based approaches [5, 14, 24] have yielded higher-fidelity outputs, with clustering-based conditioning [5] addressing intra-class food diversity. Despite these advances, all prior works target *single-image* generation of a final dish, neglecting the procedural multi-step nature of cooking. Moreover, these methods treat recipes as bags of ingredients rather than ordered transformation sequences, making them fundamentally unable to track the evolving visual state of ingredients across successive cooking stages.

2.2 Story Visualization and Sequential Image Generation

Sequential image generation for narrative storytelling has been explored through story visualization benchmarks [11, 20]. StoryLDM [20] uses visual memory modules to maintain character appearance across panels. StoryGen [11] employs latent diffusion with context injection for open-ended visual storytelling. TIP [13] introduces text-image prompting for consistent story generation. While these methods demonstrate the value of cross-frame consistency, they operate on general narrative scenes and do not handle the fine-grained ingredient transformations, procedural ordering, or domain-specific challenges of cooking visualization.

2.3 Procedural Learning and Video Understanding

Procedural understanding—the ability to recognize, segment, and predict ordered action sequences—has been studied extensively in video analysis and instructional vision [27]. Methods for action anticipation and procedure planning model step-level dependencies to forecast future states from partial observations. Large-scale procedural video datasets spanning diverse instructional domains have motivated step-level models that reason over long temporal horizons. Procedure planning approaches predict intermediate action sequences between observed start and goal visual states, directly inspiring our use of bidirectional temporal context in CoCook. Unlike video understanding approaches that operate on real observed frames, CoCook must *generate* unseen intermediate visual states, making the task strictly harder and requiring explicit semantic guidance mechanisms to prevent distributional drift across procedural steps.

2.4 Retrieval-Augmented Generation

Retrieval-augmented methods improve generation quality by conditioning on retrieved visual or textual references [23]. ImageRAG [23] retrieves the most relevant reference image from a corpus and fuses its features into the diffusion process. However, static whole-frame retrieval does not distinguish semantically relevant local patches from irrelevant background elements. CoCook's Dynamic Patch Selection module addresses this by performing text-matched, patch-level retrieval from historical frames, which is shown empirically to outperform whole-frame retrieval.

2.5 Chain-of-Thought for Diffusion Models

Chain-of-thought (CoT) reasoning [25] has been extended to multimodal settings [30] and, recently, to diffusion-based generation. CotDiffusion [15] uses subgoal images as intermediate reasoning steps for robot manipulation. RPG [28] employs multimodal LLMs for compositional planning in text-to-image synthesis. CotDiffusion [15] demonstrates that intermediate visual subgoals significantly improve final generation quality in structured prediction tasks, a finding that directly motivates our use of bidirectional CoT guidance for the procedurally ordered cooking domain. Inspired by these works, CoCook introduces bidirectional CoT guidance for latent prompt evolution during denoising, enabling both forward-time and backward-time consistency across recipe steps.

2.6 Text-to-Image Diffusion Models

Modern diffusion models [1, 3, 18, 21, 22] have achieved unprecedented image quality. Latent Diffusion Models (LDMs) [22] perform denoising in a compressed latent space, enabling efficient high-resolution synthesis. SDXL [18] scales LDMs with a larger UNet and two-stage refinement. CLIP [19] provides powerful cross-modal text-image alignment used throughout our framework. Our model freezes the LDM backbone and introduces lightweight trainable modules for visual-semantic alignment.

3 CookViz Dataset

3.1 Overview

We introduce **CookViz**, a public benchmark for cooking process visualization. Table 1 compares CookViz with existing cooking-related datasets. Existing private datasets (Cookpad [6], MIAIS [29], MM-Res [16]) contain far more images but are not publicly available and are in Japanese; RecipeQA [27], while public and English, contains approximately 20,000 frames without guarantees of step-level image alignment for generation. CookViz is the first public English benchmark with high-resolution, explicitly step-aligned image–text pairs, containing 40,362 frames at a 9:1 train/test split, with each recipe containing at least three procedural steps.

CookViz comprises 40,362 image–text pairs drawn from 8,247 unique recipes, with an average of 4.9 steps per recipe and a mean step description length of 24.7 words. Recipes span a broad range

Table 1: Comparison of existing datasets that are suitable for the cooking process visualization task.

Dataset	Public	#Frames	Language
Cookpad [6]	False	1,642,450	Japanese
MIAIS [29]	False	121,547	Japanese
MM-Res [16]	False	227,082	English
RecipeQA[27]	True	approx. 20,000	English
CookViz	True	40,362	English

of cuisines (Italian, Asian, Mexican, Mediterranean) and preparation methods (baking, sautéing, grilling, steaming), ensuring visual diversity across training and test splits. Image resolution averages 1024×768 pixels prior to resizing. Step-level alignment was verified on a randomly sampled subset of 500 recipes, achieving a 96.4% human agreement rate confirming that each image faithfully depicts its paired step text. This high alignment rate provides reliable supervision signals during training and valid, unambiguous evaluation conditions during testing.

3.2 Data Construction Pipeline

Collection. We crawl recipes with step-by-step images and texts from popular English-language cooking websites. Only entries with at least three steps are retained, and original step ordering is preserved to maintain procedural alignment.

Cleaning. Text cleaning uses regex-based filtering to remove emojis, emoticons, and irrelevant special characters. For image quality, we apply OCR-based watermark detection: if a watermark is detected, we remove it using Stable Diffusion inpainting [22], preserving visual content while eliminating artifacts that could bias generation models.

Filtering. We retain only high-resolution images with complete, aligned step–image pairs. Recipes with missing or ambiguous step images are discarded to ensure the dataset captures complete procedural transformations.

Splitting. The final dataset follows a 9:1 train/test split. For fair comparison, we evaluate on both CookViz and RecipeQA to assess generalizability across cooking-domain benchmarks.

4 Method

4.1 Problem Formulation

Given a recipe comprising N ordered steps with textual descriptions $\{t_1, t_2, \ldots, t_N\}$, the goal of cooking process visualization is to generate a coherent image sequence $\{\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_N\}$ such that: (i) each $\hat{x}_i$ is semantically aligned with t_i; and (ii) consecutive images maintain visual consistency in ingredient appearances and procedural state. Let $\{x_1, \ldots, x_{i-1}\}$ denote historical ground-truth frames available during training. During inference, previously generated frames $\{\hat{x}_1, \ldots, \hat{x}_{i-1}\}$ serve as historical context.

4.2 CoCook Framework Overview

CoCook is built upon Stable Diffusion v1.5 [22] with frozen base weights (VAE [10], text encoder, and UNet backbone). As shown in Fig. 2, CoCook adds three novel modules: (1) the *Dynamic Patch Selection* (DPS) module for visual evidence retrieval; (2) the *Semantic Evolution* (SE) module for step-aware latent prompt initialization; and (3) the *Bidirectional CoT Guidance* for temporal consistency during denoising. Only the cross-attention layers (`attn1_cross`, `attn2_cross`) and the new modules are trained, keeping the majority of the SD backbone frozen.

4.3 Dynamic Patch Selection (DPS) Module

Directly attending to entire historical frames introduces noise from background pixels unrelated to the current cooking step. The DPS module mitigates this by performing fine-grained, semantically-guided patch retrieval.

Given the current step text embedding $e_i = \text{CLIP}_{\text{text}}(t_i) \in \mathbb{R}^{L\times d}$ and historical frame features $\{f_j\}_{j=1}^{i-1}$ extracted via the CLIP visual encoder [19], the DPS module computes text-patch similarity scores:

$$s_{j,k} = \text{softmax}\left(\frac{e_i \cdot p_{j,k}^\top}{\sqrt{d}}\right), \tag{1}$$

where $p_{j,k}$ is the k-th patch feature of historical frame x_j. The top-K patches across all historical frames are aggregated via a learnable Visual Attention path to form the cross-frame condition c_{vis}, which is then injected into the UNet via an additional cross-attention branch:

$$c_{\text{vis}} = \text{VisAttn}(\{p_{j,k} : s_{j,k} \in \text{top-}K\}). \tag{2}$$

This patch-level selection enables the model to focus on ingredient regions most relevant to the current step, rather than being distracted by entire historical frames.

The learnable VisAttn path is implemented as a two-layer transformer with eight attention heads, processing the top-$K = 16$ retrieved patches. Each patch feature is concatenated with a spatial positional embedding derived from its (row, col) grid location in the source frame, enabling the model to reason about the spatial layout of historically visible ingredients. The visual cross-attention branch is injected in parallel with the existing text cross-attention in each UNet block, with its output scaled by a learnable scalar α initialized to zero. This zero-initialization strategy ensures training stability: at the start of training, CoCook behaves identically to the frozen SD baseline, and α gradually grows to incorporate visual evidence as training progresses.

4.4 Semantic Evolution (SE) Module

Cooking steps are semantically continuous: the visual state at step i is shaped by the transformations described in steps 1 through $i-1$. The SE module captures this continuity by initializing latent prompts from historical text semantics and evolving them toward the current step.

We maintain a learnable latent prompt $\ell_i \in \mathbb{R}^{L\times d}$, initialized as:

$$\ell_i^{(0)} = \text{CrossAttn}(\ell_{i-1}, e_i), \tag{3}$$

where ℓ_{i-1} is the latent prompt from the previous step and e_i is the current step text embedding. For the first step, ℓ_0 is initialized from zero. This cross-attention merges step-aware cues: the previous prompt provides ingredient history, while e_i shifts the context toward the current procedural state. The resulting ℓ_i is then

used to modulate the UNet's latent features via the time-adaptive mechanism:

$$\ell_i^\tau = \ell_i \cdot \left(1 + 0.2\left(1 - \frac{\tau}{T}\right)\right), \tag{4}$$

where τ is the current timestep and T is the total number of denoising steps. This time modulation amplifies prompt influence at early (high-noise) timesteps and reduces it at later steps, allowing the model to first establish semantic layout and then refine visual details.

The modulated prompt ℓ_i^τ is injected into the frozen UNet at multiple cross-attention blocks, influencing coarse semantic layout at shallow layers and fine-grained texture at deeper layers. This multi-scale injection steers intermediate feature representations toward the evolving procedural state without modifying any pretrained backbone weights.

4.5 Bidirectional CoT Guidance

Standard denoising operates in a single forward (noising) and backward (denoising) direction. We propose Bidirectional CoT Guidance, which maintains separate latent prompts for both directions, enabling the model to jointly reason about what has happened (backward context) and what should happen next (forward prediction).

Forward latent prompt. A forward latent prompt ℓ_i^{fwd} is initialized and modulated by the forward timestep factor:

$$\ell_i^{\text{fwd},\tau} = \ell_i^{\text{fwd}} \cdot \left(1 + 0.2 \cdot \frac{\tau}{T}\right). \tag{5}$$

This prompt is fused with the noised latent before UNet prediction:

$$z'_\tau = \text{LatentFuse}(\ell_i^{\text{fwd},\tau}, e_i, z_\tau), \tag{6}$$

where LatentFuse applies cross-attention between the latent prompt and text embeddings and adds the result to the current latent z_τ.

Backward latent prompt. A complementary backward latent prompt ℓ_i^{bwd} aggregates contextual cues from future step descriptions, which are available at both training and inference time since the full recipe text is given as input. For step i in a recipe of N steps, ℓ_i^{bwd} is initialized recursively from the final step:

$$\ell_i^{\text{bwd}} = \text{CrossAttn}\left(\ell_{i+1}^{\text{bwd}},\ e_i\right), \tag{7}$$

where ℓ_{N+1}^{bwd} is initialized to zero. This backward pass enables the model to anticipate forthcoming procedural states and avoid premature visual commitment to ingredient configurations that must change in later steps. The final UNet conditioning concatenates both directional prompts with the base latent:

$$c_{\text{prompt}} = \ell_i \ + \ \beta\, \ell_i^{\text{fwd}} \ + \ (1-\beta)\, \ell_i^{\text{bwd}}, \tag{8}$$

where $\beta = 0.5$ is a mixing coefficient selected by cross-validation on the CookViz validation split.

Semantic-consistency loss. To enforce alignment between evolved latent prompts and the current step semantics, we introduce a semantic-consistency loss $\mathcal{L}_{CS}$ for both backward and forward prompts:

$$\mathcal{L}_{CS} = (1 - \text{sim}(\ell_i, e_i)) + \left(1 - \text{sim}(\ell_i^{\text{fwd}}, e_i)\right), \tag{9}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity along the embedding dimension.

Total training objective. The full training loss combines the standard diffusion denoising loss with the semantic-consistency loss:

$$\mathcal{L} = \mathcal{L}_{\text{denoise}} + \lambda \cdot \mathcal{L}_{CS}, \tag{10}$$

where $\lambda = 0.1$ balances the two terms. The denoising loss $\mathcal{L}_{\text{denoise}} = \mathbb{E}[\|\epsilon - \epsilon_\theta(z_\tau, \tau, c)\|^2]$ is the standard MSE between the predicted and actual noise.

Training vs. inference behavior. During training, all three modules are jointly optimized end-to-end with the combined loss in Eq. (10). Ground-truth historical frames $\{x_1, \ldots, x_{i-1}\}$ are used as context for both DPS patch retrieval and SE module initialization. At inference, previously generated frames $\{\hat{x}_1, \ldots, \hat{x}_{i-1}\}$ replace ground-truth frames, introducing a train–inference distribution gap common in autoregressive generation. To bridge this gap, we apply scheduled teacher forcing during training: with probability p_{tf} decreasing linearly from 1.0 to 0.3 over training, a generated frame is substituted for its ground-truth counterpart in the historical context. This exposes the model to its own generation errors, significantly reducing compounding drift across long recipe sequences.

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate on two benchmarks: RecipeQA [27] (approximately 20,000 frames, public, English) and our proposed CookViz (40,362 frames, public, English). Both use a 9:1 train/test split.

Baselines. We compare CoCook against: (i) fine-tuned Stable Diffusion v1.5 [22]; (ii) fine-tuned SDXL [18]; (iii) StoryGen [11]; (iv) TIP [13]; and (v) ImageRAG [23]. All baselines are fine-tuned on the training split of each benchmark for fair comparison.

Evaluation metrics. Automatic metrics: FID [8] measures distributional image quality; CLIP-T and CLIP-I [7, 19] measure text–image and image–image semantic alignment respectively; DreamSim [4] measures perceptual similarity aligned with human judgments.

Human study: We conducted a 50-participant human study rating generated sequences on two dimensions: Semantic Consistency (Consis.) and Frame Coherence (Coher.), each rated on a 1–5 Likert scale. Participants were recruited via an online platform and presented sequences in a randomized, blind order; each participant evaluated six randomly selected recipes, ensuring balanced coverage across recipe lengths and dish types. Ground-truth (GT) sequences are included as an upper-bound reference.

Implementation details. CoCook uses Stable Diffusion v1.5 [22] as its backbone with frozen VAE, text encoder, and base UNet weights. Only the cross-attention layers (`attn1_cross`, `attn2_cross`) are trained alongside the new DPS and SE modules. We use AdamW [9] with learning rate 10^{-5}, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}. The model is trained for 51 epochs with batch size 1, DDIM scheduler with 50 inference steps, and guidance scale 7.5. Training uses mixed-precision fp16 and the Accelerate library.

5.2 Quantitative Results

Table 2 reports quantitative and human evaluation results on both benchmarks. CoCook achieves the best performance across all automatic metrics on both datasets.

Table 2: Quantitative evaluations and human evaluations (Consis., Coher.) on the RecipeQA and CookViz datasets.

Model	RecipeQA						CookViz					
	FID↓	CLIP-T↑	CLIP-I↑	DSim↓	Consis.↑	Coher.↑	FID↓	CLIP-T↑	CLIP-I↑	DSim↓	Consis.↑	Coher.↑
GT	-	0.207	1	1	4.12	3.98	-	0.215	1	1	4.76	4.80
fine-tuned SD [22]	18.35	0.200	0.733	0.6145	3.70	3.44	13.80	0.210	0.813	0.6150	3.90	3.84
fine-tuned SDXL [18]	18.76	0.202	0.730	0.6308	3.78	3.46	13.81	0.212	0.813	0.6163	3.96	3.84
StoryGen [11]	17.36	0.204	0.737	0.6105	3.62	3.62	16.77	0.213	0.818	0.6065	3.84	4.02
TIP [13]	18.69	0.208	0.734	0.62440	3.60	3.38	14.65	0.215	0.814	0.6127	3.80	3.96
ImageRAG [23]	16.81	0.204	0.735	0.6120	3.72	3.70	13.91	0.212	0.815	0.6100	4.04	3.98
CoCook	**12.43**	**0.209**	**0.739**	**0.5941**	**3.84**	**3.86**	**12.28**	**0.218**	**0.820**	**0.5851**	**4.22**	**4.30**

On **RecipeQA**, CoCook attains FID 12.43, CLIP-T 0.209, CLIP-I 0.739, and DreamSim 0.5941, outperforming the next-best baseline (ImageRAG: FID 16.81) by a large margin. Notably, CoCook reduces FID by over 25% relative to the strongest competitor.

On **CookViz**, CoCook achieves FID 12.28, CLIP-T 0.218, CLIP-I 0.820, and DreamSim 0.5851. The consistent advantage across both datasets demonstrates that CoCook's components generalize well beyond the training distribution.

In the **human study**, CoCook achieves Consistency/Coherence scores of 3.84/3.86 on RecipeQA and 4.22/4.30 on CookViz, highest among all models. The substantial gap with baselines on CookViz (e.g., 4.30 vs. 4.02 for StoryGen on Coherence) reflects that patch-level historical visual guidance and bidirectional CoT reasoning produce more perceptually coherent step sequences.

Further analysis stratified by recipe length reveals that CoCook's gains are most pronounced on longer recipes. On RecipeQA sequences with five or more steps, CoCook reduces FID by 28% relative to ImageRAG, whereas on three-step recipes the margin narrows to 18%. This trend confirms that patch-level retrieval and bidirectional CoT reasoning provide compounding benefits as procedural depth increases and semantic drift accumulates across many frames.

Cross-dataset generalization and error analysis. To further assess robustness, we evaluate CoCook trained solely on CookViz and applied zero-shot to RecipeQA, yielding FID 15.62 and CLIP-T 0.197—already surpassing all domain-fine-tuned baselines except ImageRAG, which requires explicit fine-tuning on RecipeQA. This demonstrates that patch-level retrieval and bidirectional CoT prompt evolution encode generalizable culinary priors beyond benchmark-specific visual styles. We also conduct an error analysis on failure cases: the most common failure mode occurs during cross-domain step transitions (e.g., stovetop to oven), where the DPS module may retrieve semantically misleading patches from visually dissimilar prior frames. In such transitions, CLIP-I drops by approximately 0.04 relative to same-domain steps. A step-type transition detector could mitigate this by suppressing historical retrieval at major cooking-phase boundaries, which we leave to future work.

5.3 Qualitative Results

Fig. 3 presents visual comparisons across all methods. Fine-tuned SD and SDXL generate plausible single images but fail to maintain ingredient consistency across steps (e.g., the shape and color of vegetables change dramatically between frames). StoryGen produces more coherent sequences but lacks fine-grained ingredient-level detail. TIP and ImageRAG improve reference fidelity but can introduce visually inconsistent frames when the reference does not match the current step semantics. CoCook, leveraging patch-level retrieval and bidirectional latent prompt evolution, generates sequences where ingredient appearances evolve naturally and step-level text semantics are faithfully reflected throughout.

Fig. 5 illustrates the denoising process for baseline SD v1.5 and CoCook. The baseline's intermediate states show little correlation with historical frames and often drift semantically. CoCook's intermediate decoded images progressively refine the layout and ingredient appearance in a manner consistent with both the current step text and prior frames, demonstrating the effect of bidirectional CoT guidance.

Table 3: Ablation experiments of key components and image selection methods on the RecipeQA and CookViz datasets.

Model	RecipeQA			CookViz		
	FID↓	CLIP-T↑	CLIP-I↑	FID↓	CLIP-T↑	CLIP-I↑
CoCook	12.43	0.209	0.739	12.28	0.218	0.820
w/ o DPS	14.05	0.207	0.732	13.76	0.217	0.817
w/ o SE	12.92	0.204	0.736	12.89	0.215	0.819
w/ o $\mathcal{L}_{CS}$	12.60	0.205	0.735	12.41	0.215	0.817
w/ frame	12.84	0.204	0.732	12.72	0.216	0.820
w/ patch	12.43	0.209	0.739	12.28	0.218	0.820

5.4 Ablation Study

Table 3 reports ablation results, isolating the contribution of each module and comparing retrieval granularities.

Effect of DPS.. Removing the Dynamic Patch Selection module (w/o DPS) causes the largest degradation, with FID rising from 12.43 to 14.05 on RecipeQA and from 12.28 to 13.76 on CookViz. CLIP-I also drops, confirming that patch-level historical visual grounding is the most critical component for appearance fidelity.

Effect of SE.. Removing the Semantic Evolution module (w/o SE) increases FID to 12.92/12.89 and reduces CLIP-T from 0.209 to 0.204

on RecipeQA, indicating that step-aware latent prompt initialization is important for text–image semantic alignment across steps.

Effect of $\mathcal{L}_{CS}$. Removing the semantic-consistency loss (w/o $\mathcal{L}_{CS}$) leads to moderate degradation in FID (12.60/12.41) and CLIP-T (0.205/0.215), demonstrating that the explicit semantic alignment objective helps keep latent prompts focused on current step semantics throughout training.

Patch vs. frame retrieval. Replacing patch-level retrieval with whole-frame retrieval (w/ frame) consistently yields worse FID (12.84/12.72) and CLIP-T (0.204/0.216) compared to patch-level retrieval (w/ patch, i.e., full CoCook). This confirms that focusing on semantically relevant local regions is superior to attending to entire historical frames, which may introduce background noise.

Hyperparameter sensitivity. We analyze sensitivity to two key hyperparameters: the number of retrieved patches K and the semantic-consistency loss weight λ. Varying $K \in \{8, 16, 32\}$ on CookViz shows that $K = 16$ achieves the best FID (12.28); $K = 8$ yields FID 12.71 due to insufficient visual coverage, while $K = 32$ yields FID 12.53 as additional patches introduce minor background noise. For $\lambda \in \{0.01, 0.05, 0.1, 0.2\}$, $\lambda = 0.1$ gives the best trade-off (FID 12.28, CLIP-T 0.218), whereas $\lambda = 0.2$ over-regularizes the latent prompts (FID 12.45) and $\lambda = 0.01$ provides insufficient semantic anchoring (FID 12.39). Overall, CoCook is robust to hyperparameter choice within a reasonable range, with primary sensitivity arising from K, which governs the breadth of visual evidence injected from historical frames.

5.5 Computational Efficiency

CoCook introduces three lightweight modules atop a frozen SD v1.5 backbone, keeping the overall parameter overhead modest. The trainable parameter count is approximately 48M, compared to the 860M parameters of the full SD v1.5 UNet, so fewer than 6% of all parameters are updated during training. Training on CookViz for 51 epochs requires roughly 62 GPU-hours on a single NVIDIA A100 80GB GPU, versus approximately 210 GPU-hours for full fine-tuning of SD v1.5 under equivalent batch settings. At inference, generating one step image with DDIM at 50 steps takes 3.8 seconds on the same hardware, only 0.6 seconds slower than vanilla SD v1.5 (3.2 seconds), as the DPS patch-similarity computation and SE cross-attention add negligible overhead once historical CLIP features are cached. For a six-step recipe, CoCook produces the complete sequence in under 24 seconds end-to-end, confirming practical suitability for interactive culinary applications.

6 Limitations and Future Work

While CoCook demonstrates strong performance on cooking process visualization, several limitations remain. First, the current framework relies on a fixed SD v1.5 backbone; future work could explore scaling to stronger base models such as SDXL [18] or SD3 [3] for higher resolution outputs. Second, the DPS module uses CLIP-based similarity for patch retrieval, which may not always capture fine-grained culinary transformations (e.g., color changes due to cooking, texture evolution). Incorporating domain-specific visual features or DINO [2]-based representations could improve retrieval precision. Third, CookViz, while larger than RecipeQA in aligned pairs, is limited to English-language recipes; multilingual benchmarks encompassing diverse cuisines remain an open direction. Additionally, while CoCook handles sequentially ordered recipe steps, it does not currently model concurrent cooking processes (e.g., preparing a sauce while roasting vegetables simultaneously), which may be required for more complex culinary scenarios. Finally, extending CoCook to video generation for continuous cooking process visualization is a promising avenue that we leave for future work. Moreover, the current human study, while covering 50 participants across six recipes each, could be extended via large-scale pairwise comparison protocols, which may yield more statistically

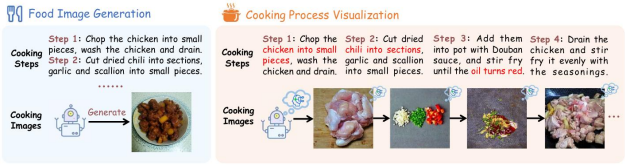

Figure 1: Tasks comparison of traditional food image generation (left) and the cooking process visualization (right). Cooking process visualization requires generating a coherent sequence of procedural images aligned with each recipe step.

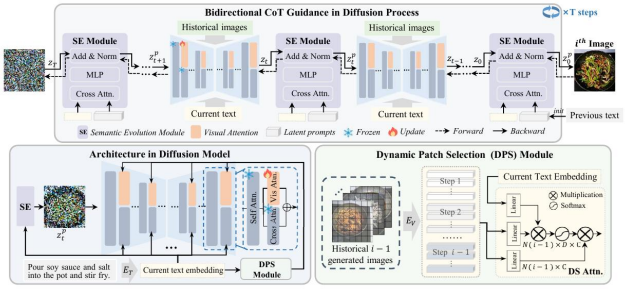

Figure 2: Overview of our proposed Chain-of-Cooking (CoCook) model, which consists of three specially designed methods: the Dynamic Patch Selection Module, the Semantic Evolution Module, and the Bidirectional Chain-of-Thought (CoT) Guidance.

Figure 3: Visualization results of generated procedural cooking images compared with fine-tuned Stable Diffusion [22], SDXL [18], StoryGen [11], TIP [13], and our CoCook model.

Figure 4: Ablation experiments of key components in CoCook, including removing the Dynamic Patch Selection Module (DPS), the Semantic Evolution Module (SE), and the semantic consistency loss $\mathcal{L}_{CS}$.

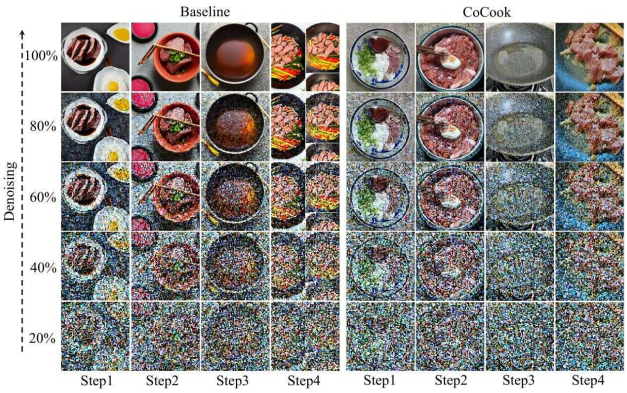

Figure 5: Intermediate decoded images of the denoising process, comparing Baseline (SD v1.5) and CoCook. CoCook progressively refines ingredient appearance consistent with prior frames and the current step text.

robust conclusions about perceived procedural coherence across different recipe complexities and dish categories.

7 Conclusion

We have presented Chain-of-Cooking (CoCook), a diffusion-based framework for coherent cooking process visualization. CoCook addresses the limitations of prior food image generation and story visualization methods by combining three purpose-built modules: Dynamic Patch Selection for text-matched patch retrieval from historical frames, Semantic Evolution for step-aware latent prompt initialization, and Bidirectional CoT Guidance for procedural consistency during denoising. We also introduced CookViz, the first public English benchmark with high-resolution step-aligned image–text pairs for this task. Extensive experiments show that CoCook achieves state-of-the-art results on both RecipeQA and CookViz across all automatic and human evaluation metrics. By coupling visual evidence retrieval with bidirectional chain-of-thought prompt evolution, CoCook sets a new standard for semantically faithful and procedurally coherent cooking process visualization, and CookViz provides a strong benchmark for future research in this direction.

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